Support Vector Machines — Detailed Dry Runs and Computations

November 6, 2025

Contents

1 Overview

This document collects step-by-step derivations and numerical "dry runs" for linear Support Vector Machines (SVMs). It contains:

- Margin derivation and geometric interpretation
- Hard-margin SVM (primal) with a complete 1D numerical example
- Dual formulation and solution for the same example
- Soft-margin SVM formulation and numeric example with slack variables
- Kernel remark and small feature map example
- A short sklearn example showing SVM in practice

2 Notation and Setup

We consider training data $\{(x_i, y_i)\}_{i=1}^{R}$ where $x_i \in \mathbb{R}^m$ and $y_i \in \{+1, -1\}$. The linear classifier is:

$$f(x; w, b) = \text{sign}(w \cdot x + b)$$

Margin planes are chosen as:

$$\text{plus-plane: } w \cdot x + b = +1, \qquad \text{minus-plane: } w \cdot x + b = -1$$

3 Margin Width Derivation

Let x_+ be on the plus-plane and x_- on the minus-plane. The vector w is perpendicular to these planes. Assume $x_+ = x_- + \lambda w$.

Using the plane equations:

$$\begin{aligned} w \cdot x_+ + b &= 1 \\ w \cdot x_- + b &= -1 \end{aligned}$$

Subtracting gives:

$$w \cdot (x_+ - x_-) = 2$$

Substitute $x_+ - x_- = \lambda w$:

$$\lambda\, w \cdot w = 2 \quad \Rightarrow \quad \lambda = \frac{2}{\|w\|^2}$$

Hence margin width:

$$M = \|x_+ - x_-\| = |\lambda|\, \|w\| = \frac{2}{\|w\|}$$

Maximizing M is equivalent to minimizing $\frac{1}{2}\|w\|^2$.

4 Hard-Margin SVM Primal

For linearly separable data:

$$\begin{aligned}
&\min_{w,b} \frac{1}{2}\|w\|^2 \\
&\text{s.t. } y_i(w \cdot x_i + b) \geq 1, \quad i = 1, \dots, R
\end{aligned} \tag{1}$$

5 1D Hard-Margin Example

5.1 Problem Setup

Let:

$$x_1 = 2,\ y_1 = +1; \qquad x_2 = -2,\ y_2 = -1$$

We solve the primal problem (1).

5.2 Constraints

$$2w + b \geq 1 \tag{2}$$
$$2w - b \geq 1 \tag{3}$$

At optimum both are tight:

$$\begin{aligned}
2w + b &= 1 \\
2w - b &= 1
\end{aligned}$$

Add:

$$4w = 2 \Rightarrow w = \frac{1}{2}$$

Substitute:

$$2(0.5) + b = 1 \Rightarrow b = 0$$

5.3 Margin and Objective

$$\|w\| = 0.5 \quad \Rightarrow \quad M = \frac{2}{0.5} = 4$$

$$\text{Objective } = \frac{1}{2}w^2 = \frac{1}{8} = 0.125$$

Classifier: $\text{sign}(0.5x)$.

6 Dual Formulation

The Lagrangian:

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2}\|w\|^2 - \sum_i \alpha_i [y_i(w \cdot x_i + b) - 1]$$

Stationarity:

$$w = \sum_i \alpha_i y_i x_i \tag{4}$$

$$0 = \sum_i \alpha_i y_i \tag{5}$$

Dual:

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

s.t. $\alpha_i \geq 0$, $\sum_i \alpha_i y_i = 0$.

6.1 Numeric Dual Solution

Using $x_1 = 2$, $x_2 = -2$, $y_1 = 1$, $y_2 = -1$:

$$D(\alpha) = \alpha_1 + \alpha_2 - \frac{1}{2}(4\alpha_1^2 + 8\alpha_1\alpha_2 + 4\alpha_2^2)$$

Constraint: $\alpha_1 = \alpha_2 = \alpha$.

$$D(\alpha) = 2\alpha - 8\alpha^2$$

Derivative:

$$\frac{dD}{d\alpha} = 2 - 16\alpha = 0 \Rightarrow \alpha = \frac{1}{8}$$

Then:

$$w = \sum_i \alpha_i y_i x_i = 4\alpha = \frac{1}{2}, \quad b = 0$$

7 Soft-Margin SVM

Allow violations via slack variables $\xi_i \geq 0$:

$$\begin{aligned} &\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_i \xi_i \\ &\text{s.t. } y_i(w \cdot x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \end{aligned} \tag{6}$$

7.1 1D Example

Let:

$$x_1 = 0.8,\ y_1 = +1; \quad x_2 = -2,\ y_2 = -1$$

Constraints:

$$0.8w + b + \xi_1 = 1 \tag{7}$$

$$2w - b = 1 \tag{8}$$

Combine:

$$2.8w + \xi_1 = 2 \Rightarrow \xi_1 = 2 - 2.8w$$

Objective:

$$J(w) = \frac{1}{2}w^2 + C(2 - 2.8w)$$

Derivative:

$$J'(w) = w - 2.8C = 0 \Rightarrow w = 2.8C$$

Feasibility: $\xi_1 = 2 - 7.84C \geq 0 \Rightarrow C \leq 0.255$.

8 KKT Conditions

- $w = \sum_i \alpha_i y_i x_i$
- $\alpha_i[y_i(w \cdot x_i + b) - 1 + \xi_i] = 0$
- $0 < \alpha_i < C \Rightarrow$ on margin; $\alpha_i = C \Rightarrow$ inside margin or misclassified

9 Understanding Slack Variables and the Role of C

9.1 Why We Need Slack Variables

In real life, data is often **not perfectly separable**. A few points might lie on the wrong side of the separating line (or plane). To handle such imperfect data, SVM introduces **slack variables** ξ_i (Greek letter xi).

Each ξ_i tells us:

- $\xi_i = 0 \rightarrow$ the point is correctly classified and outside or on the margin.
- $0 < \xi_i < 1 \rightarrow$ the point is inside the margin but still correctly classified.
- $\xi_i > 1 \rightarrow$ the point is misclassified (on the wrong side of the decision boundary).

So, the **soft-margin SVM** allows some margin violations by paying a penalty for each $\xi_i > 0$.

9.2 The Role of the Parameter C

The constant $C > 0$ in the soft-margin SVM objective:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_i \xi_i$$

controls how much we **penalize violations**.

- A **large** C means: We strongly penalize margin violations. The model tries very hard to classify every training point correctly (less tolerant to errors). This may cause the model to **overfit** (fit training data too tightly).
- A **small** C means: We allow more violations (more flexible boundary). The model focuses more on having a wider margin, even if some points are misclassified. This can help with **generalization** (better test performance).

9.3 Geometric Intuition

- The first term $\frac{1}{2}\|w\|^2$ wants a **large margin** (small w).
- The second term $C\sum_i \xi_i$ wants **few or small violations**.

C controls the trade-off:

$$\text{Large margin (simple model)} \quad \leftrightarrow \quad \text{Low training error (complex model).}$$

9.4 Example Illustration

If $C = 0.1$:

$$J(w) = \frac{1}{2}w^2 + 0.1(2 - 2.8w)$$

$\rightarrow$ smaller C gives a smaller weight to violations, allowing some slack.
If $C = 5$:

$$J(w) = \frac{1}{2}w^2 + 5(2 - 2.8w)$$

$\rightarrow$ larger C forces the model to reduce ξ_i as much as possible.

9.5 Summary Table

Parameter	Effect on Model	Risk
Small C	Wide margin, more tolerance to errors	May underfit
Large C	Narrow margin, less tolerance to errors	May overfit

9.6 Beginner Tip

Think of C like a teacher's *strictness*:

- If C is high $\rightarrow$ teacher is strict $\rightarrow$ every mistake (misclassified point) gets punished $\rightarrow$ perfect marks needed.
- If C is low $\rightarrow$ teacher is lenient $\rightarrow$ a few mistakes are okay $\rightarrow$ broader learning.

Choosing C is about finding the right balance between **accuracy** and **robustness**.

10 Kernel Trick

Replace dot product with kernel $K(x_i, x_j)$:

$$K(a, b) = \phi(a) \cdot \phi(b)$$

Common kernels:

- Polynomial: $(a \cdot b + c)^d$
- RBF: $\exp(-\gamma\|a - b\|^2)$
- Sigmoid: $\tanh(\kappa a \cdot b + \theta)$

11 Practical Example (Python)

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.svm import SVC
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

X = np.array([[2,2],[1,1],[-1,-1],[-2,-2]])
y = np.array([1,1,-1,-1])

model = make_pipeline(StandardScaler(), SVC(kernel='linear', C=1.0))
model.fit(X, y)

svc = model.named_steps['svc']
print("w =", svc.coef_[0], "b =", svc.intercept_[0])
```

12 Closing Notes

This document shows all algebraic steps and numeric results for small examples to visualize the role of C, w, b, and α. Next steps can include adding a 2D numeric case with plots or embedding visual decision boundaries directly in the PDF.